POWER UP THE BRAIN AND GET INTO THE ZONE:

We're hard wired at birth to react to stress in certain physical ways. Now our ancient instincts, honed over millennia to survive the attack of a sabre-toothed tiger or fight off raiders from another tribe, could be managed to produce better rugby players according to Colquhoun, A. (2007). We are born with an unlimited amount of brain potential. The main functions of the brain are thinking, learning and creativity. "This makes brain power the number one determiner for success" Rugby News Annual (2005). The sports person who has the ability to out-think, out-wit, out-learn, out-create and out-play the competition will have a definite advantage.

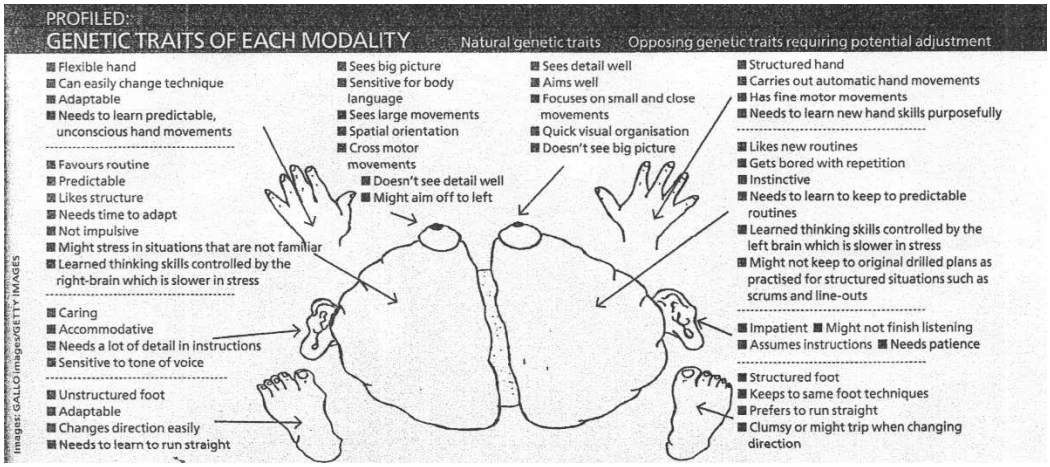
Brain Fitness:
We are all born with two brain hemispheres, but we don't automatically use both of them. This is referred to as **homolateral**, which means we only utilise one hemisphere of the brain at a time in alternating fashion. Homolaterality slows people's learning, thinking and creative capacity down, resulting in them being more frustrated, having to work longer and harder, rather than faster and smarter according to Vermeulen, A. (2005). Utilising both hemispheres comes through constant development and stimulation. The good news however is that all people can be **bilaterial**, engaging both brain hemispheres simultaneously provided people are exposed to brain-based development interventions and brain integrated techniques and activities. Using both the left and right brain hemispheres will enable you to work faster and smarter and therefore make learning easier.

The level of brain fitness is a very important aspect impacting the performance of rugby players, because utilising both brain hemispheres simultaneously will be a vital pre-requisite to out-think, out-wit, out-learn, out-create and out-play their competition on the rugby field according to Vermeulen (2005). Research indicates that the brain can become unfit in a period of as short as five days without any stimulation. According to Rugby News Annual (2005), players who study or practice other professional careers perform better than professional rugby players who do nothing else. "It is clear that new challenges and brain-based stimulation grows balanced rugby players who perform better on the sports field" Vermeulen (2005).

Brain Profiling:
Brain profiling predicts where our mental and physical strengths and weaknesses lie and, most importantly, how we will function under stress. According to Colquhoun (2007), we are genetically programmed at conception to have five dominant sensory perceptions: Brain hemisphere, eye, ear, hand and foot. This is why we instinctively favour one hand over the other or step on or off with a certain foot. It is the same with our eyes and ears; we're either **left-brain dominant** (predisposed to routine and structure) or **right-brain dominant** (bored by repetition). The scary thing is that one can actually find out which players have big match temperament and who don't.

Genetic brain profiling plays an important role in predicting what happens to a player when he gets tired and put under pressure. According to Lotter, "in simple terms, what happens when we're under stress or tired is that we begin to lose access to the non-dominant hemisphere and the modalities functioning from it". It is further stated that if this stress intensifies we start to see the same loss in the dominant hemisphere. For example, if it is the **right hemisphere**, the ability to plan and forward think would be impaired, and if it is in the **left hemisphere**, the ability to perform repetitive functions would suffer.

Right-brain players might be best suited for to the decision-making positions whereas those positions with more repetitive techniques may be better suited to **left-brain** dominances according to Lotter.



<u>Warming up the brain</u>:
As much as the body has to be warmed up to ensure peak physical performance, so the brain should be "switched on" and warmed up if a rugby player wants to out-play the opposition according to Rugby News Annual (2005). This is mandatory, but something that is very seldomly done by sports teams, because they do not know how to switch on the brain, how to activate their senses and how to switch on their neurological system for peak performance.

When a rugby player experiences too much stress, he may secrete inhibiting chemicals like **cortisol** and **morphine** that slow down electrical transmission between brain cells and may even shut down certain areas of the brain according to Rugby News Annual 2005. Vermeulen (2005), states that this is why many rugby players make stereo type mistakes under stress.

The good news is that many physical and mental actions can be taken to ensure that rugby players do not "switch off" certain brain areas under pressure. Players need to stay in the "zone" where they can optimise their brain power, concentration and physical performance.

When a rugby player thinks **negative thoughts** and experiences negative emotions he will secrete more inhibiting chemicals like **cortisol**, which will inhibit and slow down electrical transmissions between cells according to Rugby News Annual (2005). This will have a direct impact on the rugby players' performance resulting in him making mistakes or wrong tactical decisions. These inhibitors also weaken the immune system, drain energy levels, thus leading to ill health and bad physical performance.

If a player thinks **positive thoughts** and experiences constructive emotions he will secrete facilitating chemicals like **seratonin** and **endorphins** that will transmit electrical messages between cells, strengthening their immune system, enhancing mental activity and increasing their performance. All rugby players should be equipped with mental re-patterning skills to ensure that they replace negative mindsets with positive ones according to Rugby News Annual (2005). To help ensure the success of a team the players must have confidence and a constructive approach.

Getting in the Zone:

The brain goes into different states. Every brain state has a different function. When a rugby player runs onto the field, it does not mean that his brain will just automatically be in the right state for peak performance, if he did not take action to activate the desired brain state according to Rugby News Annual (2005). The ideal brain state is a state of relaxed alertness, a state where he can think and perform well. According to Rugby News Annual (2005) this is referred to being in the "zone" where rugby players can be focussed, highly motivated and think on their feet well without stress letting them lose control.

One of the most powerful terms in world sport today is the mental state called "the zone" according to Townsend, C. (2007). This is the mental state that produces super-human performances, amazing times and winning streaks. "An unusual feeling of effortless power, allowing them to feel as if their body is powered by a turbo-charged engine, and they find their body moving with maximum result but minimum energy" Townsend (2007). When an athlete reaches this state they often feel that they are not controlling it themselves and that they are not responsible for the performance. A common question by an athlete would be "how did I do that?" So, what exactly is it? This powerful force which guides the body during this period comes from the subconscious mind. The source of all our bodily movement and past sporting memories and experience come from our subconscious.

When a player is in the "zone" their conscious mind becomes quiet and this allows the more powerful subconscious mind to take over. The athlete is then in "automatic pilot", in the way a computer runs software. This allows their movements to flow much easier, effortlessly increasing speed and power in a way that could never be matched by the conscious thought. "This means that when you in the zone, you have virtually no thought going through your mind whatsoever, your body is just moving on instinct and automatic-pilot, powered by your subconscious mind" Townsend (2007).

Every person is unique and they have their own unique way of getting into the zone. Being in the "zone" is a state of where you are focused, relaxed, and self-trusting according to Rana, S (2007). The superego referred to as the "inner voice" prevents you from getting into the "zone" because it works on you by making you doubt anything and everything and in turn reducing your self-worth. Rana (2007) states that to get into the "zone" you have to shut off the Critical Advisor (super ego) which has a critical mentality. In a team sport like rugby the player needs to learn to forget about the result, but focus more on their own performance. A person in the "zone" only focuses on the present and will not allow any negative emotions from the past or future effect their performance.

Unique brain-based strengths:
Rugby players also need to know what their unique brain-based strengths and talents are. According to Rugby News Annual (2005) there are four categories of intelligence and eleven different forms of intelligence. All people are smart but, in different ways. If a player knows how he is smart it will help him outplay the opposition.

A good captain should have strong **emotional intelligence** which implies that he can control himself well under pressure and will still be able to healthily interact with team mates and make smart decisions on the field. Also being strong language smart will contribute to increased effectiveness of communication on the sports field according to Rugby News Annual (2005).

A "game breaker" will have strong **creative intelligence** as this capacity will enable him to make a quick strategic analysis of the game and take actions to out-wit his opposition. Creative intelligence is normally a prerequisite for back line players and some loose forwards according to Rugby Annual News (2005).

People become successful at what they do when they primarily focus on enhancing and utilising their strengths. Knowing one's strengths will give you an indication of the gifts and talents you have to be able to out-play the competition.

If a player is **right brain dominant** it would make that person a conceptual and strategic thinker on the field. This player would be able to grasp the bigger picture of the situation on the field at once, making a creative decision on how to out-play their opponents. The right brain dominant player will also see links between other sports and rugby.

Another area to look at is the players **learning receptivity**. If this is high the player will learn from other people and experiences easily, be mentally flexible to change, and adapt easily to new situations like rules that change or new environments.

If a player is **logical / mathematical intelligent** they will focus on acquiring and practicing the specific techniques required for their position and analyse the details of the technique.

A strong **visual / spatial intelligence** allows a player to use the space that he has on the field more creatively because they can picture and use the space instantly.

Strong **intrapersonal intelligence** makes a player accurately aware of his own strengths and weaknesses and allows him to manage himself well. A player with this strength will not lose focus easily or be easily intimidated.

Some players may also show a strong preference for **physical intelligence**, which implies that they enjoy acquiring and executing physical skills.

Strong **spiritual intelligence** will make a player very purpose driven and passionate.

Linguistic intelligence allows a player to efficiently and clearly communicate with his players.

A player who shows a greater **intrapersonal (self) intelligence** may be more individualistic than a player who shows a greater **interpersonal (people) intelligence**. A player with stronger interpersonal intelligence will be a far better team player.

Having poor **musical intelligence** may cause a player to lose rhythm when under pressure. For example a hooker or kicker may lose rhythm under pressure if his musical intelligence was poor.

Conclusion:
Everyone is unique and has been genetically programmed to react to different situations in their own way, in order to reach their full potential. There is substantial evidence from this article that your performance will increase from knowing who you are and what your strengths or weaknesses are. If the individual in a team can learn how to reach their optimal performance level, collectively the team's performance will be unrivalled by a team who has not been able to. Get rid of that "inner voice" let your subconscious take control and get into the "zone"!

References:

Colquhoun, A. (2007). GENETIC IMPERATIVES. Popular Mechanics, September 2007.

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MULTIPLE INTELLIGENCES:

Some of the ways of exploring and solving problems listed in this study may feel uncomfortable to you while others seem comfortable. That's because you do have a preferred, personal learning style. Everyone in the team has their own strengths. This is why it is important to know what they are. Some players may be suited to specific positions more than others. This is the aim of this exercise.

Think of these intelligences as members of your team, various approaches will be most suitable in specific areas. Look to maximise your team's performance by being open to various intelligences.

Here is a brief summary of some of the different types of intelligences:

1. **LINGUISTIC**: Good with language; can debate effectively; able to write easily; avid reader… A good captain in a rugby team should score high in this intelligence. **Linguistic intelligence** allows a player to efficiently and clearly communicate with his players.
1. **MATHEMATICAL LOGIC**: Effective in describing patterns; can generate mathematical formulae; able to design / conduct an experiment… If a player is **logical / mathematical intelligent** they will focus on acquiring and practicing the specific techniques required for their position and analyse the details of the technique.
2. **BODILY / PHYSICAL**: Enjoy building / constructing; enjoy hands-on opportunities; enjoy the outdoors… Some players may also show a strong preference for **physical intelligence**, which implies that they enjoy acquiring and executing physical skills.
3. **VISUAL / SPATIAL**: Invent board games; can draw illustrate, sketch etc.; able to use a map effectively; effectively use maps, graphs, diagrams… A strong **visual / spatial intelligence** allows a player to use the space that he has on the field more creatively because they can picture and use the space instantly.
4. **MUSICAL**: Naturally pick up rhythms; enjoy whistling / singing; enjoy explaining how song is made up with various instruments…If this is poor it may cause a player to lose rhythm when under pressure e.g. a hooker or kicker.
5. **INTER-PERSONAL**: Gifted with social skills; able to impart your knowledge; able to conduct meetings / discussions; socially gregarious…A player with stronger **interpersonal (people) intelligence** will be a far better team player.
6. **INTRA-PERSONAL**: Able to describe your strengths, set & pursue a goal; describe your personal values… Strong **intrapersonal (self) intelligence** makes a player accurately aware of his own strengths and weaknesses and allows him to manage himself well. A player with this strength will not lose focus easily or be easily intimidated.
7. **NATURALIST**: Cares for pets, insects…; use microscopes, telescopes…; notice the natural environment; keen observation skills…

Multiple Intelligence Self Inventory

Read the statements listed below. Decide how strongly you agree / disagree that each statement best represents you. Use the scale:

1 = strongly disagree 2 = disagree 3 = uncertain 4 = agree 5 = strongly agree

1	Books are important to me.	1	2	3	4	5
2	I can easily compute numbers in my head.	1	2	3	4	5
3	I am sensitive to colour.	1	2	3	4	5
4	I engage in at least one sport or physical activity on a regular basis.	1	2	3	4	5
5	I have a pleasant singing voice.	1	2	3	4	5
6	I am the sort of person people come to for advice and counsel.	1	2	3	4	5
7	I regularly spend time alone meditating, reflecting or thinking about important life questions.	1	2	3	4	5
8	I enjoy being outdoors.	1	2	3	4	5
9	When I drive on the freeway, I pay attention to the words on the advertising boards than the scenery.	1	2	3	4	5
10	I enjoy playing games or solving brain teasers that require logical thinking.	1	2	3	4	5
11	I prefer looking at reading material that is heavily illustrated.	1	2	3	4	5
12	I find it difficult to sit still for long periods of time.	1	2	3	4	5
13	I know the tunes to many different songs or musical pieces.	1	2	3	4	5
14	I consider myself as a leader.	1	2	3	4	5
15	I am able to respond to setbacks with resilience.	1	2	3	4	5
16	I enjoy classifying rocks or plants or animals.	1	2	3	4	5
17	I enjoy word games like Scrabble and cross-words.	1	2	3	4	5
18	I believe that almost everything has a rational explanation.	1	2	3	4	5
19	I have vivid dreams at night.	1	2	3	4	5
20	I can mimic and imitate others well.	1	2	3	4	5
21	I ‘hear’ melodies, rhythms and harmonies in my mind.	1	2	3	4	5
22	I would rather spend my evenings at a lively party than stay at home alone.	1	2	3	4	5
23	I have a special hobby or interest that pretty much keeps me to myself.	1	2	3	4	5
24	I accept the effectiveness of homeopathy.	1	2	3	4	5
25	My conversation includes frequent reference to things that I have read or heard.	1	2	3	4	5
26	I like finding logical flaws in things that people say and do at home and work.	1	2	3	4	5
27	I like to draw or doodle.	1	2	3	4	5
28	I need to touch things in order to learn more about them.	1	2	3	4	5
29	I can tell when a musical note is off-key.	1	2	3	4	5
30	I touch others while talking to them.	1	2	3	4	5
31	I consider myself to be strong-willed or independent minded.	1	2	3	4	5
32	I look for patterns in nature.	1	2	3	4	5
33	I can persuade and convince others with words.	1	2	3	4	5
34	I feel content when testing hypotheses using measurements.	1	2	3	4	5
35	I can ‘picture’ where all the furniture will fit in a new place.	1	2	3	4	5
36	I would describe myself as well co-ordinated.	1	2	3	4	5
37	I often make tapping sounds or sing little melodies while working.	1	2	3	4	5
38	I enjoy performing tasks with other people.	1	2	3	4	5
39	I can detect and express complex feelings in myself.	1	2	3	4	5
40	I am fascinated by the weather.	1	2	3	4	5

Self Inventory – Score Sheet

	L	ML	VS	BP	M	Inter	Intra	N
	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24
	25	26	27	28	29	30	31	32
	33	34	35	36	37	38	39	40
Totals:								

These intelligences can be used as a guideline for position specific characteristics. A coach could use this test to help him make a decision on which player to select.